

DATA BASE SYSTEMS ARCHITECTURE

Databases III

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 DataBases Infrastructure 
- 2 DBMS Architecture 
- 3 Database System Administration 
- 4 Transactional System 
- 5 Query Execution 
- 6 Concurrency Control 
- 7 Failure Recovery 



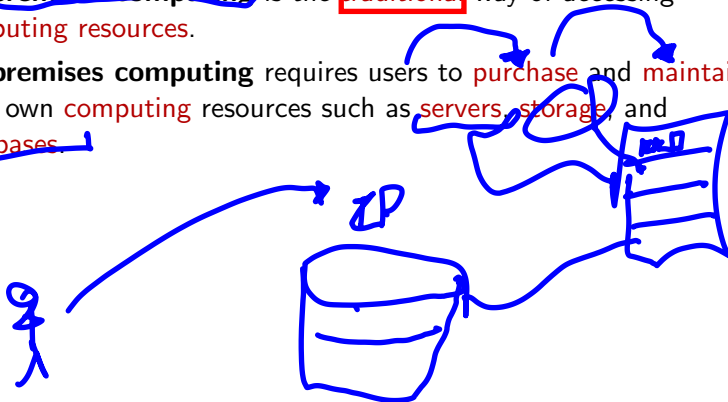
Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery



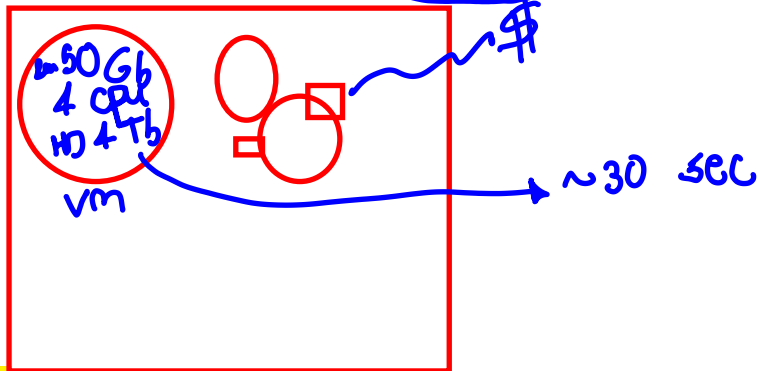
What is On-Premises Computing?

- **On-premises computing** is the **traditional** way of accessing computing resources.
- **On-premises computing** requires users to **purchase** and **maintain** their own **computing** resources such as **servers**, **storage**, and **databases**.



What is the Cloud Computing?

- **Cloud computing** is the delivery of **computing services** over the internet.
- **Cloud computing** allows users to **access computing resources** such as servers, storage, and databases **on demand**.

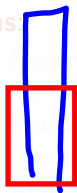


Pros & Cons of Cloud Computing

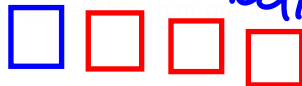
● Pros:

- **Cost-Effective:** Cloud computing is a cost-effective way to access computing resources.
- **Scalable:** Cloud computing is a scalable way to access computing resources.
- **Flexible:** Cloud computing is a flexible way to access computing resources.

● Cons:



vertical



horizontal



Pros & Cons of Cloud Computing

● Pros:

- **Cost-Effective:** Cloud computing is a **cost-effective** way to access computing resources.
- **Scalable:** Cloud computing is a **scalable** way to access computing resources.
- **Flexible:** Cloud computing is a **flexible** way to access computing resources.

● Cons:

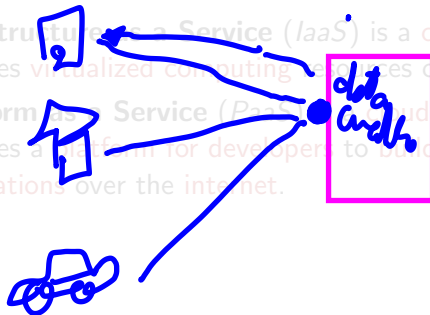
- **Security:** Cloud computing can be **less secure** than on-premises computing.
- **Performance:** Cloud computing can be **slower** than on-premises computing.
- **Reliability:** Cloud computing can be **less reliable** than on-premises computing.

network



SaaS Vs. IaaS Vs. PaaS

- **Software as a Service (SaaS)** is a **software distribution** model in which a **third-party** provider **hosts applications** and makes them available to customers over the **internet**.



- **Infrastructure as a Service (IaaS)** is a cloud computing model that provides **virtualized computing resources** over the internet.
- **Platform as a Service (PaaS)** is a cloud computing model that provides a **platform for developers** to **build, deploy, and manage applications** over the internet.



SaaS Vs. IaaS Vs. PaaS

- **Software as a Service (SaaS)** is a **software distribution** model in which a **third-party** provider **hosts applications** and makes them available to customers over the **internet**.
- **Infrastructure as a Service (IaaS)** is a **cloud computing** model that provides **virtualized computing** resources over the **internet**.
- **Platform as a Service (PaaS)** is a **cloud computing** model that provides a **platform** for developers to build, deploy, and manage applications over the internet.

zac → terraform

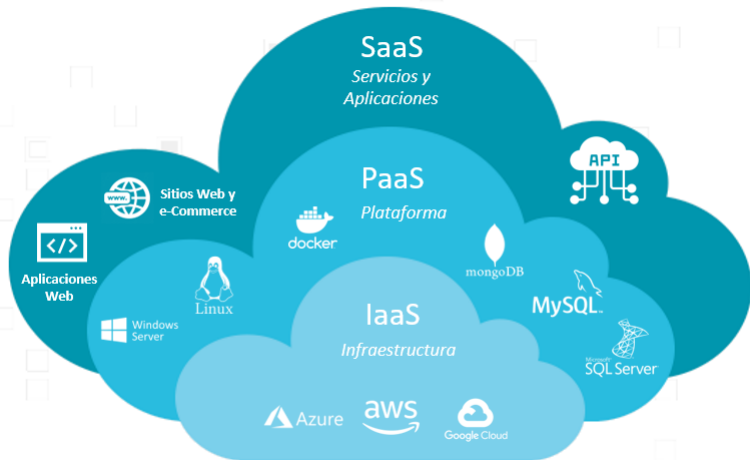


SaaS Vs. IaaS Vs. PaaS

- **Software as a Service** (*SaaS*) is a **software distribution** model in which a **third-party** provider **hosts applications** and makes them available to customers over the **internet**.
- **Infrastructure as a Service** (*IaaS*) is a **cloud computing** model that provides **virtualized computing** resources over the **internet**.
- **Platform as a Service** (*PaaS*) is a **cloud computing** model that provides a **platform for developers** to **build**, **deploy**, and **manage** applications over the **internet**.



Cloud Levels

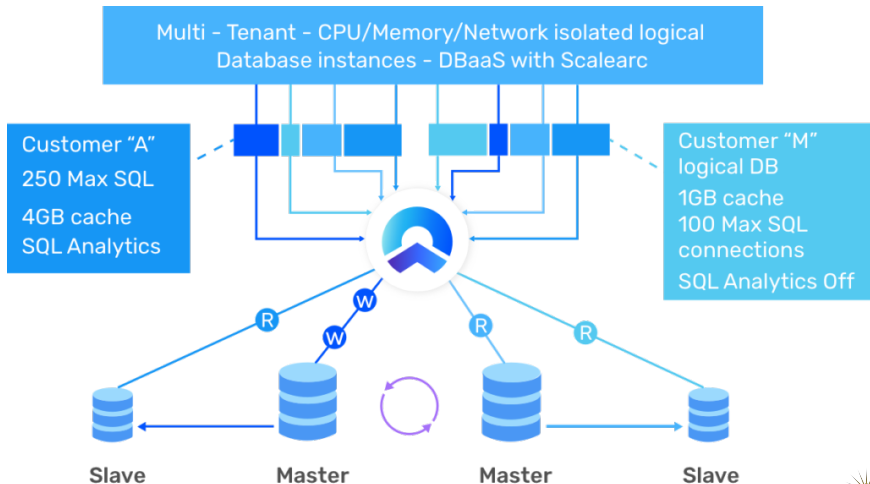


DataBases as a Service

Database as a Service (DBaaS) is a **cloud computing model** that provides **database services** over the **internet**.



Case Study: DBaaS Custom for Clients



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture**
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.
- Each component is responsible for a **specific aspect** of data management.



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.
- Each component is responsible for a **specific aspect** of data management.



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.
- Each component is responsible for a **specific aspect** of data management.



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.
- Each component is responsible for a **specific aspect** of data management.



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.
- Each component is responsible for a **specific aspect** of data management.



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.
- Each component is responsible for a **specific aspect** of data management.

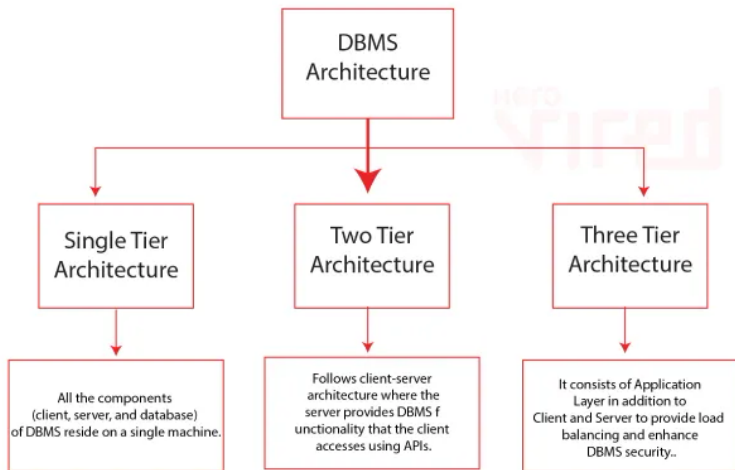


DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.
- Each component is responsible for a **specific aspect** of data management.



DBMS Architecture Tiers



DBMS Architecture N-Tier

DBMS Architecture



DatabaseTown.com

It is responsible for providing an interface for users.

It is responsible for managing the different views of the data in the database.

It is responsible for managing the logical organization of data in the database.

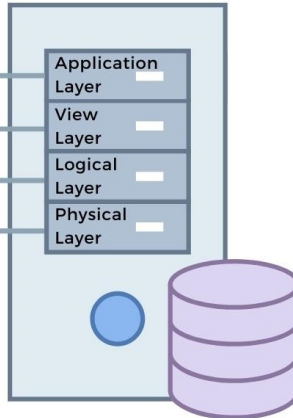
It is responsible for managing the physical storage of data on disk.

Application Layer

View Layer

Logical Layer

Physical Layer



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: **Combines** features of centralized and distributed systems.
- **Peer-to-Peer DBMS**: Each node can act as a **client** and **server**.
- **In memory DBMS**: Data is stored in **RAM** for faster access.



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: **Combines** features of centralized and distributed systems.
- **Peer-to-Peer DBMS**: Each node can act as a **client** and **server**.
- **In memory DBMS**: Data is stored in **RAM** for faster access.



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: **Combines** features of centralized and distributed systems.
- **Peer-to-Peer DBMS**: Each node can act as a **client** and **server**.
- **In memory DBMS**: Data is stored in **RAM** for faster access.



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: **Combines** features of centralized and distributed systems.
- **Peer-to-Peer DBMS**: Each node can act as a **client** and **server**.
- **In memory DBMS**: Data is stored in **RAM** for faster access.



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: **Combines** features of centralized and distributed **systems**.
- **Peer-to-Peer DBMS**: Each node can act as a **client** and **server**.
- **In memory DBMS**: Data is stored in **RAM** for faster access.



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: **Combines** features of centralized and distributed **systems**.
- **Peer-to-Peer DBMS**: Each node can act as a **client** and **server**.
- **In memory DBMS**: Data is stored in **RAM** for faster access.



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: **Combines** features of centralized and distributed **systems**.
- **Peer-to-Peer DBMS**: Each node can act as a **client** and **server**.
- **In memory DBMS**: Data is stored in **RAM** for faster access.



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration**
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - Data storage management
 - System architecture design
- Database administrators (DBAs) are **essential** for modern organizations.



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - **Data storage management**
 - System architecture design
 - Transaction and query processing
 - Concurrency control
 - Backup and recovery
 - Security and access control
- **Database administrators (DBAs)** are **essential** for modern organizations.



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - **Data storage management**
 - **System architecture design**
 - Transaction and query processing
 - Concurrency control
 - Backup and recovery
 - Security and access control
- **Database administrators (DBAs)** are **essential** for modern organizations.



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - Data storage management
 - System architecture design
 - Transaction and query processing
 - Concurrency control
 - Backup and recovery
 - Security and access control
- Database administrators (DBAs) are essential for modern organizations.



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - Data storage management
 - System architecture design
 - Transaction and query processing
 - Concurrency control
 - Backup and recovery
 - Security and access control
- Database administrators (DBAs) are essential for modern organizations.



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - Data storage management
 - System architecture design
 - Transaction and query processing
 - Concurrency control
 - Backup and recovery
 - Security and access control
- Database administrators (DBAs) are **essential** for modern organizations.



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - Data storage management
 - System architecture design
 - Transaction and query processing
 - Concurrency control
 - Backup and recovery
 - Security and access control
- Database administrators (DBAs) are **essential** for modern organizations.



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - Data storage management
 - System architecture design
 - Transaction and query processing
 - Concurrency control
 - Backup and recovery
 - Security and access control
- **Database administrators (DBAs)** are **essential** for modern organizations.



Record Storage Concepts

- A **record** (or row/tuple) is the **basic unit** of data storage in a *database table*.
- **Efficient record storage** is crucial for **fast data retrieval** and **update**.
- **Storage techniques:**
 - **Heap storage:** Unordered, fast inserts.
 - **Clustered storage:** Ordered, fast range queries.
 - **Indexed storage:** Uses indexes for fast lookups.



Record Storage Concepts

- A **record** (or row/tuple) is the **basic unit** of data storage in a *database table*.
- **Efficient record storage** is crucial for **fast data retrieval** and **update**.
- Storage techniques:
 - Heap storage: Unordered, fast inserts.
 - Sequential storage: Ordered, fast range queries.
 - Indexed storage: Uses indexes for fast lookups.



Record Storage Concepts

- A **record** (or row/tuple) is the **basic unit** of data storage in a *database table*.
- **Efficient record storage** is crucial for **fast data retrieval** and **update**.
- Storage techniques:
 - **Heap storage**: Unordered, fast inserts.
 - **Sequential storage**: Ordered, fast range queries.
 - **Indexed storage**: Uses indexes for fast lookups.



Record Storage Concepts

- A **record** (or row/tuple) is the **basic unit** of data storage in a *database table*.
- **Efficient record storage** is crucial for **fast data retrieval** and **update**.
- Storage techniques:
 - **Heap storage**: Unordered, fast inserts.
 - **Sequential storage**: Ordered, fast range queries.
 - **Indexed storage**: Uses indexes for fast lookups.



Record Storage Concepts

- A **record** (or row/tuple) is the **basic unit** of data storage in a *database table*.
- **Efficient record storage** is crucial for **fast data retrieval** and **update**.
- Storage techniques:
 - **Heap storage**: Unordered, fast inserts.
 - **Sequential storage**: Ordered, fast range queries.
 - **Indexed storage**: Uses indexes for fast lookups.



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.
- Block size and layout affect I/O performance.
- Records may be packed, slotted, or may span multiple blocks.
- Free space management is important for updates and inserts.
- Fragmentation management ensures efficient use of space.
- Data compression can also improve storage efficiency.



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.
- **Block size** and **layout** affect **I/O performance**.
- Records may be **packed**, **slotted**, or may **span multiple blocks**.
- **Free space management** is important for **updates** and **inserts**.
- **Fragmentation management** ensures efficient use of space.
- **Data compression** can also improve storage efficiency.



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.
- **Block size** and **layout** affect **I/O performance**.
- Records may be **packed**, **slotted**, or may **span multiple blocks**.
- Free space management is important for **updates** and **inserts**.
- Fragmentation management ensures efficient use of space.
- Data compression can also improve storage efficiency.



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.
- **Block size** and **layout** affect **I/O performance**.
- Records may be **packed**, **slotted**, or may **span multiple blocks**.
- **Free space management** is important for **updates** and **inserts**.
- **Fragmentation management** ensures efficient use of space.
- **Data compression** can also improve storage efficiency.



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.
- **Block size** and **layout** affect **I/O performance**.
- Records may be **packed**, **slotted**, or may **span multiple blocks**.
- **Free space management** is important for **updates** and **inserts**.
- **Fragmentation management** ensures efficient use of space.
- **Data compression** can also improve storage efficiency.



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.
- **Block size** and **layout** affect **I/O performance**.
- Records may be **packed**, **slotted**, or may **span multiple blocks**.
- **Free space management** is important for **updates** and **inserts**.
- **Fragmentation management** ensures efficient use of space.
- **Data compression** can also improve storage efficiency.



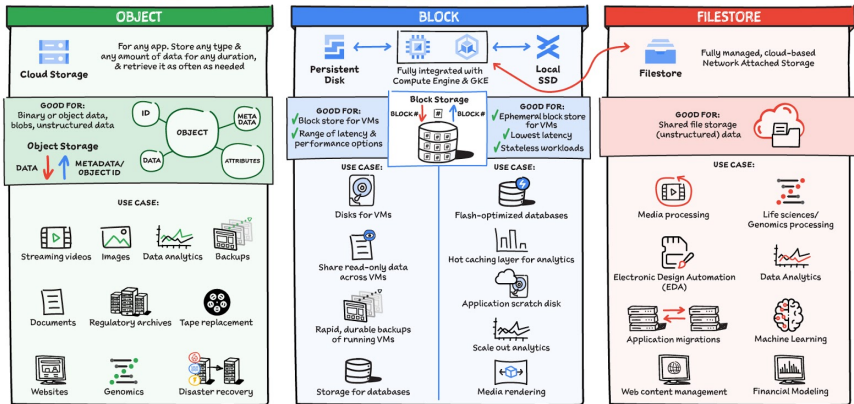
Record Storage: Use Cases

#GCPsketchnote

@PVERGADIA
THECLOUDGIRL.DEV
04.23.2021



Which Storage Should I Use?



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System**
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery



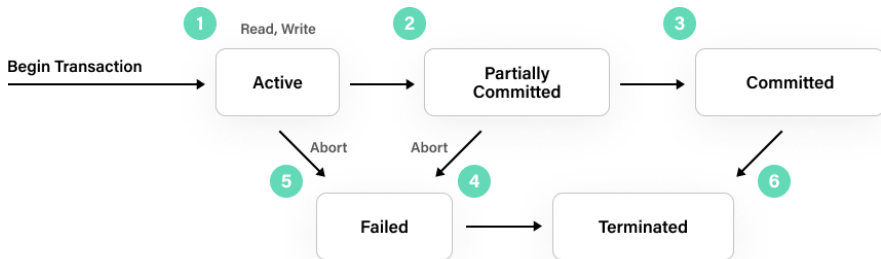
Transactional System Concepts

- A **transaction** is a sequence of operations performed as a **single logical unit of work**.
- Transactions must satisfy the **ACID** properties:
 - **Atomicity**: All or nothing.
 - **Consistency**: Preserves database integrity.
 - **Isolation**: Transactions do not interfere.
 - **Durability**: Results persist after completion.



Transaction Lifecycle

- **Begin**: Transaction starts.
- **Read/Write**: Operations are performed.
- **Commit**: Changes are made permanent.
- **Rollback**: Changes are undone if an error occurs.
- **Savepoints** can be used for partial rollbacks.



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution**
- 6 Concurrency Control
- 7 Failure Recovery

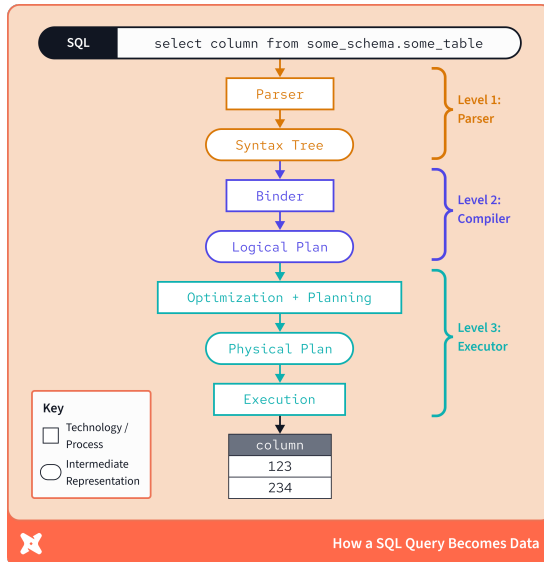


Query Execution Process

- **Query execution** is the process of **interpreting** and **running** database queries.
- Steps:
 - **Parsing**: Analyzing query syntax.
 - **Optimization**: Choosing the best execution plan.
 - **Execution**: Retrieving and processing data.
- **Efficient execution** is critical for **performance**.



Query Execution Flow: Full Transaction



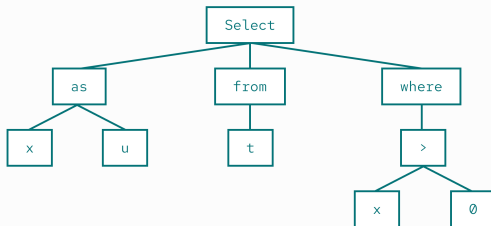
Query Execution Flow: Syntax Tree

SQL

`select x as u from t where x > 0`

Parser

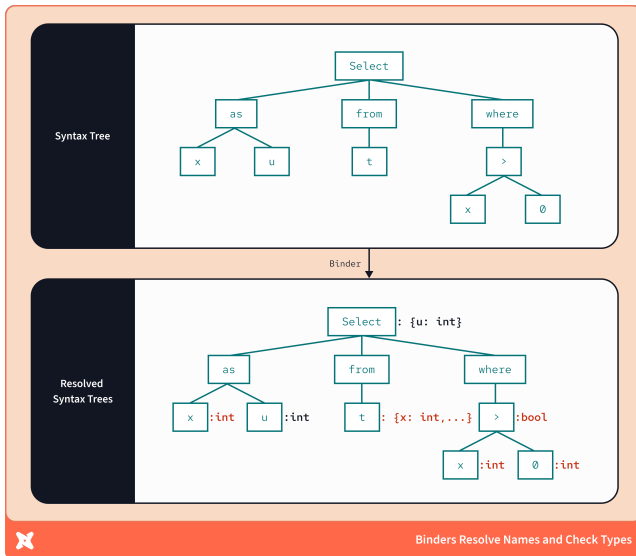
Syntax Tree



Parsers Recognize the Structure of the Query

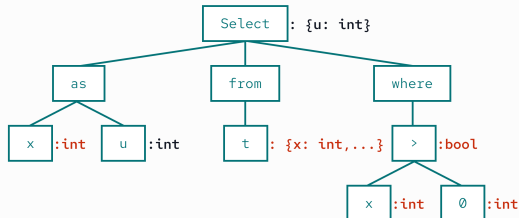


Query Execution Flow: Compilation



Query Execution Flow: Logical Plan

Resolved
Syntax Trees



Generating Logical Plan

Logical Plan

1. TableScan: t
2. Filter: t.x > 0
3. Projection: t.x as u



Compilation Produces an Executable Plan from a Resolved Syntax Tree



Query Optimization

- The **query optimizer** selects the most efficient strategy for executing a query.
- Considers indexes, join methods, and data distribution.
- May rewrite queries for better performance.
- Cost-based and rule-based optimization approaches.



Query Optimization

- The **query optimizer** selects the most efficient strategy for executing a query.
- Considers **indexes**, **join methods**, and **data distribution**.
- May **rewrite queries** for better performance.
- **Cost-based** and **rule-based** optimization approaches.



Query Optimization

- The **query optimizer** selects the most efficient strategy for executing a query.
- Considers **indexes**, **join methods**, and **data distribution**.
- May **rewrite queries** for better performance.
- **Cost-based** and **rule-based** optimization approaches.

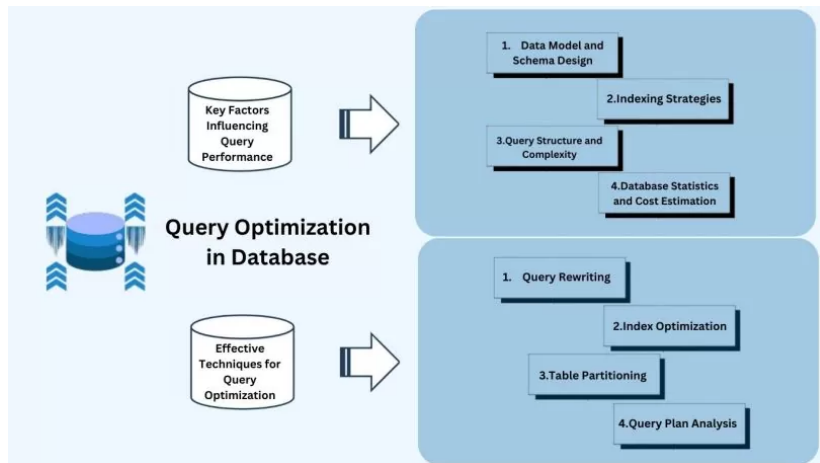


Query Optimization

- The **query optimizer** selects the most efficient strategy for executing a query.
- Considers **indexes**, **join methods**, and **data distribution**.
- May **rewrite queries** for better performance.
- **Cost-based** and **rule-based** optimization approaches.



Query Optimization Factors



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control**
- 7 Failure Recovery



Concurrency Control Concepts

- **Concurrency control** ensures correct results when **multiple transactions** execute **simultaneously**.
- Prevents problems like **lost updates**, **dirty reads**, and **deadlocks**.
- Techniques:
 - Locking protocols: Use locks to control access (e.g., two-phase locking).
 - Timestamp ordering: Assigns timestamps to transactions.
 - Optimistic concurrency control: Assumes no conflicts.
 - Multiversion concurrency control: Assumes conflicts may occur.



Concurrency Control Concepts

- **Concurrency control** ensures correct results when **multiple transactions** execute **simultaneously**.
- Prevents problems like **lost updates**, **dirty reads**, and **deadlocks**.
- Techniques:
 - **Locking protocols**: Use locks to control access (e.g., two-phase locking).
 - **Timestamp ordering**: Assigns timestamps to transactions.
 - **Optimistic concurrency control**: Assumes no conflicts.
 - **Pessimistic concurrency control**: Assumes conflicts may occur.



Concurrency Control Concepts

- **Concurrency control** ensures correct results when **multiple transactions** execute **simultaneously**.
- Prevents problems like **lost updates**, **dirty reads**, and **deadlocks**.
- Techniques:
 - **Locking protocols**: Use locks to control access (e.g., two-phase locking).
 - **Timestamp ordering**: Assigns timestamps to transactions.
 - **Optimistic concurrency control**: Assumes no conflicts.
 - **Pessimistic concurrency control**: Assumes conflicts may occur.



Concurrency Control Concepts

- **Concurrency control** ensures correct results when **multiple transactions** execute **simultaneously**.
- Prevents problems like **lost updates**, **dirty reads**, and **deadlocks**.
- Techniques:
 - **Locking protocols**: Use locks to control access (e.g., two-phase locking).
 - **Timestamp ordering**: Assigns timestamps to transactions.
 - **Optimistic concurrency control**: Assumes no conflicts.
 - **Pessimistic concurrency control**: Assumes conflicts may occur.



Concurrency Control Concepts

- **Concurrency control** ensures correct results when **multiple transactions** execute **simultaneously**.
- Prevents problems like **lost updates**, **dirty reads**, and **deadlocks**.
- Techniques:
 - **Locking protocols**: Use locks to control access (e.g., two-phase locking).
 - **Timestamp ordering**: Assigns timestamps to transactions.
 - **Optimistic concurrency control**: Assumes no conflicts.
 - **Pessimistic concurrency control**: Assumes conflicts may occur.



Locking and Deadlocks

- **Locks** prevent **conflicting operations** on the same data.
- **Deadlocks** occur when transactions **wait indefinitely** for each other.
- DBMS uses **deadlock detection** and **resolution strategies**.
- **Isolation levels** (e.g., Read Committed, Serializable) control visibility of changes.



Locking and Deadlocks

- **Locks** prevent **conflicting operations** on the same data.
- **Deadlocks** occur when transactions **wait indefinitely** for each other.
- DBMS uses **deadlock detection** and **resolution strategies**.
- **Isolation levels** (e.g., **Read Committed**, **Serializable**) control visibility of changes.



Locking and Deadlocks

- **Locks** prevent **conflicting operations** on the same data.
- **Deadlocks** occur when transactions **wait indefinitely** for each other.
- DBMS uses **deadlock detection** and **resolution strategies**.
- **Isolation levels** (e.g., **Read Committed**, **Serializable**) control visibility of changes.



Locking and Deadlocks

- **Locks** prevent **conflicting operations** on the same data.
- **Deadlocks** occur when transactions **wait indefinitely** for each other.
- DBMS uses **deadlock detection** and **resolution strategies**.
- **Isolation levels** (e.g., **Read Committed**, **Serializable**) control visibility of changes.



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery**



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.
- Types of failures:
 - Transaction failure: Only one transaction fails.
 - System crash: All active transactions are lost.
 - Media failure: Disk or storage device fails.
- Recovery techniques:
 - Write-ahead logging (WAL):
 - Log created before writing data.
 - Ensures durability and atomicity.
 - Can be used for recovery.
 - Periodically update the database state.
 - Recovery is easy when a failure occurs.
 - Can be used for recovery of data pages.
 - Always recover to a consistent state.



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.
- Types of failures:
 - **Transaction failure**: Only one transaction fails.
 - **System crash**: All active transactions are lost.
 - **Media failure**: Disk or storage device fails.
- Recovery techniques:
 - Write-ahead logging (WAL):
 - Two-phase commit
 - Checkpointing
 - Periodically saving the database state



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.
- Types of failures:
 - **Transaction failure**: Only one transaction fails.
 - **System crash**: All active transactions are lost.
 - **Media failure**: Disk or storage device fails.
- Recovery techniques:
 - **Write-ahead logging (WAL)**:
 - Logs changes before applying them.
 - Ensures durability and atomicity.
 - **Checkpointing**:
 - Periodically saves the database state.
 - Reduces recovery time.
 - **Shadow paging**:
 - Maintains copies of data pages.
 - Allows quick recovery to a consistent state.



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.
- Types of failures:
 - **Transaction failure**: Only one transaction fails.
 - **System crash**: All active transactions are lost.
 - **Media failure**: Disk or storage device fails.
- Recovery techniques:
 - **Write-ahead logging (WAL)**:
 - Logs changes before applying them.
 - Ensures durability and atomicity.
 - **Checkpointing**:
 - Periodically saves the database state.
 - Reduces recovery time.
 - **Shadow paging**:
 - Maintains copies of data pages.
 - Allows quick recovery to a consistent state.



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.
- Types of failures:
 - **Transaction failure**: Only one transaction fails.
 - **System crash**: All active transactions are lost.
 - **Media failure**: Disk or storage device fails.
- Recovery techniques:
 - **Write-ahead logging (WAL)**:
 - Logs changes before applying them.
 - Ensures durability and atomicity.
 - **Checkpointing**:
 - Periodically saves the database state.
 - Reduces recovery time.
 - **Shadow paging**:
 - Maintains copies of data pages.
 - Allows quick recovery to a consistent state.



Write-Ahead Logging and Checkpoints

- **Write-ahead logging (WAL):** All changes are **logged before** being applied to the database.
- **Checkpoints:** Periodically **save** the database **state** to speed up recovery.
- **Shadow paging:** Maintains **copies** of data **pages**.
- **Recovery** ensures atomicity and durability.



Write-Ahead Logging and Checkpoints

- **Write-ahead logging (WAL):** All changes are **logged before** being applied to the database.
- **Checkpoints:** Periodically **save** the database **state** to speed up recovery.
- **Shadow paging:** Maintains **copies** of data **pages**.
- **Recovery** ensures atomicity and durability.



Write-Ahead Logging and Checkpoints

- **Write-ahead logging (WAL):** All changes are **logged before** being applied to the database.
- **Checkpoints:** Periodically **save** the database **state** to speed up recovery.
- **Shadow paging:** Maintains **copies** of data **pages**.
- Recovery ensures atomicity and durability.



Write-Ahead Logging and Checkpoints

- **Write-ahead logging (WAL):** All changes are **logged before** being applied to the database.
- **Checkpoints:** Periodically **save** the database **state** to speed up recovery.
- **Shadow paging:** Maintains **copies** of data **pages**.
- **Recovery ensures atomicity and durability.**



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/databases-ii>

